

Advanced GC-Orbitrap-HRMS method for the simultaneous multi-class quantification of priority and emerging contaminants in thermally processed paprika and almonds

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Posters del grupo QAC

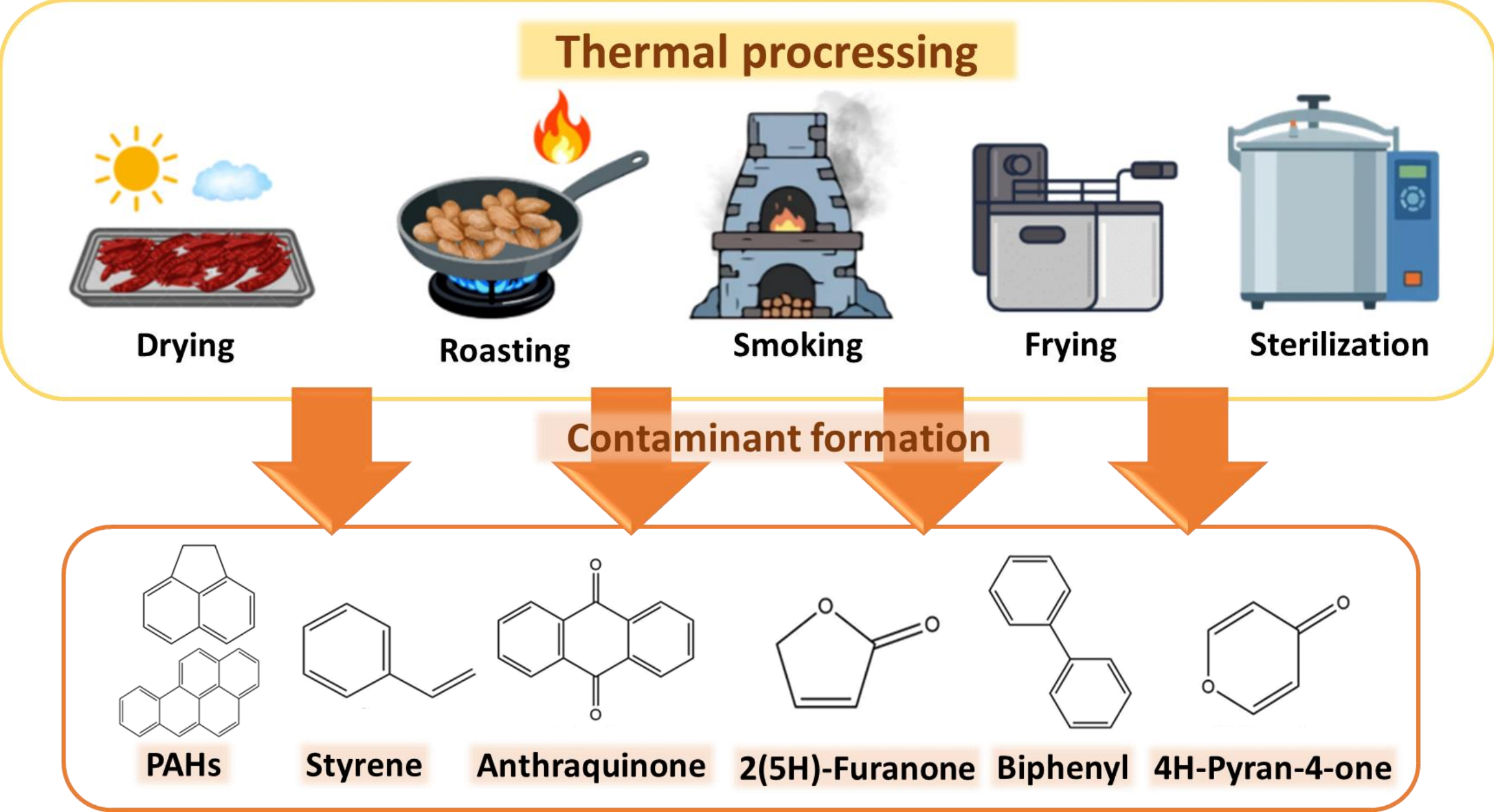
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INTRODUCTION

During the thermal treatment of food, such as **paprika** smoking or **almond** frying and roasting, a variety of compounds may be generated through multiple chemical pathways. Recent studies have emphasised the complexity of thermally induced reactions and the simultaneous formation of multiple classes of volatile and **contaminant compounds** in food matrices. This study [1] investigates the formation of process-induced contaminants across various thermal treatments in paprika (dried, sterilised and smoked) and almond (raw, roasted and fried). A total of **25 analytes were simultaneously determined**. These included 15 PAHs, styrene, and anthraquinone, classified as hazardous contaminants by the International Agency for Research on Cancer (IARC), alongside emerging thermal markers, specifically 6 furanone derivatives, 1 pyranone derivative, and benzofuran. A **novel multicomponent analytical** method based on ultrasound-assisted extraction coupled with gas chromatography and Orbitrap high-resolution mass spectrometry (UAE-GC-Orbitrap-HRMS) was developed and validated. To the best of our knowledge, **this is the first study reporting the simultaneous determination of a broad range of thermal process-induced compounds** (both priority and emerging) in these food matrices.

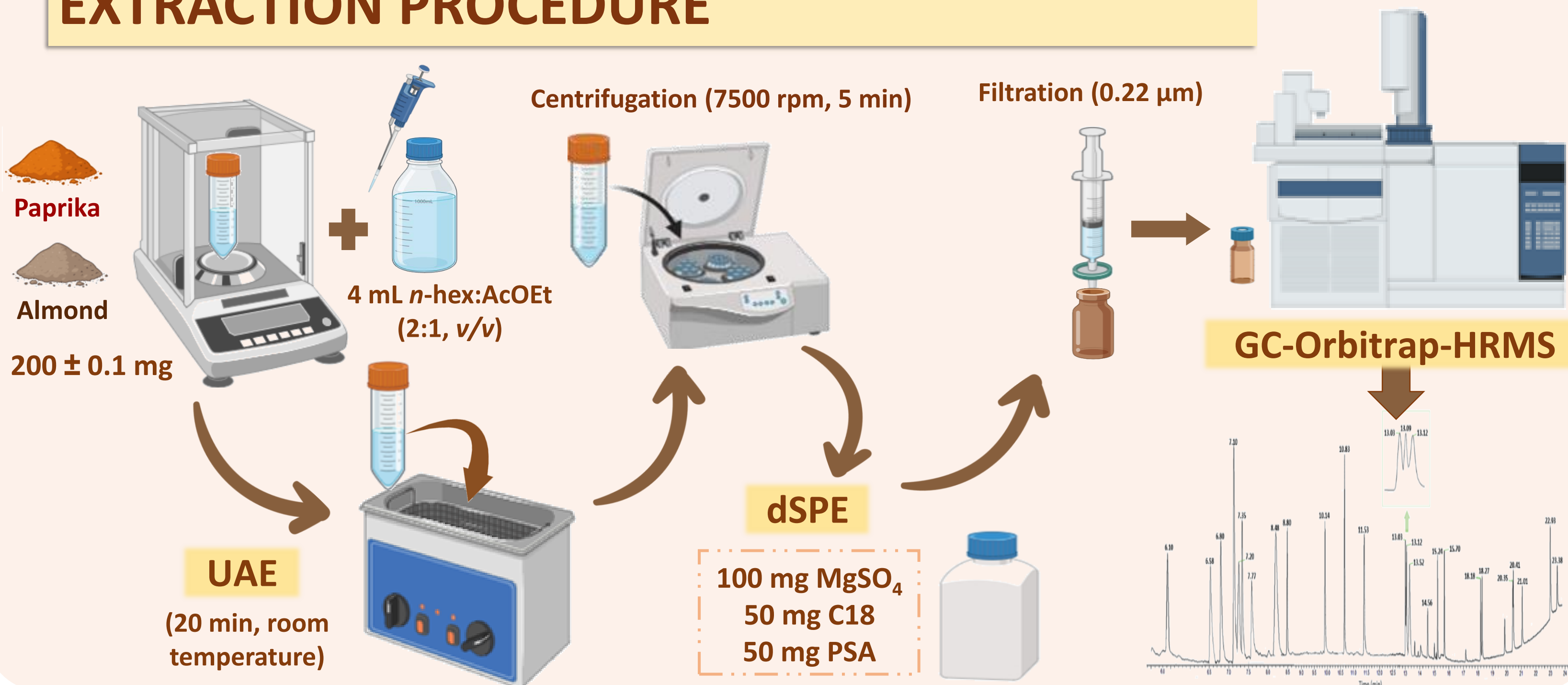


SAMPLES

Twelve **paprika** samples and **twelve almond** samples subjected to different thermal processing conditions were analysed



EXTRACTION PROCEDURE

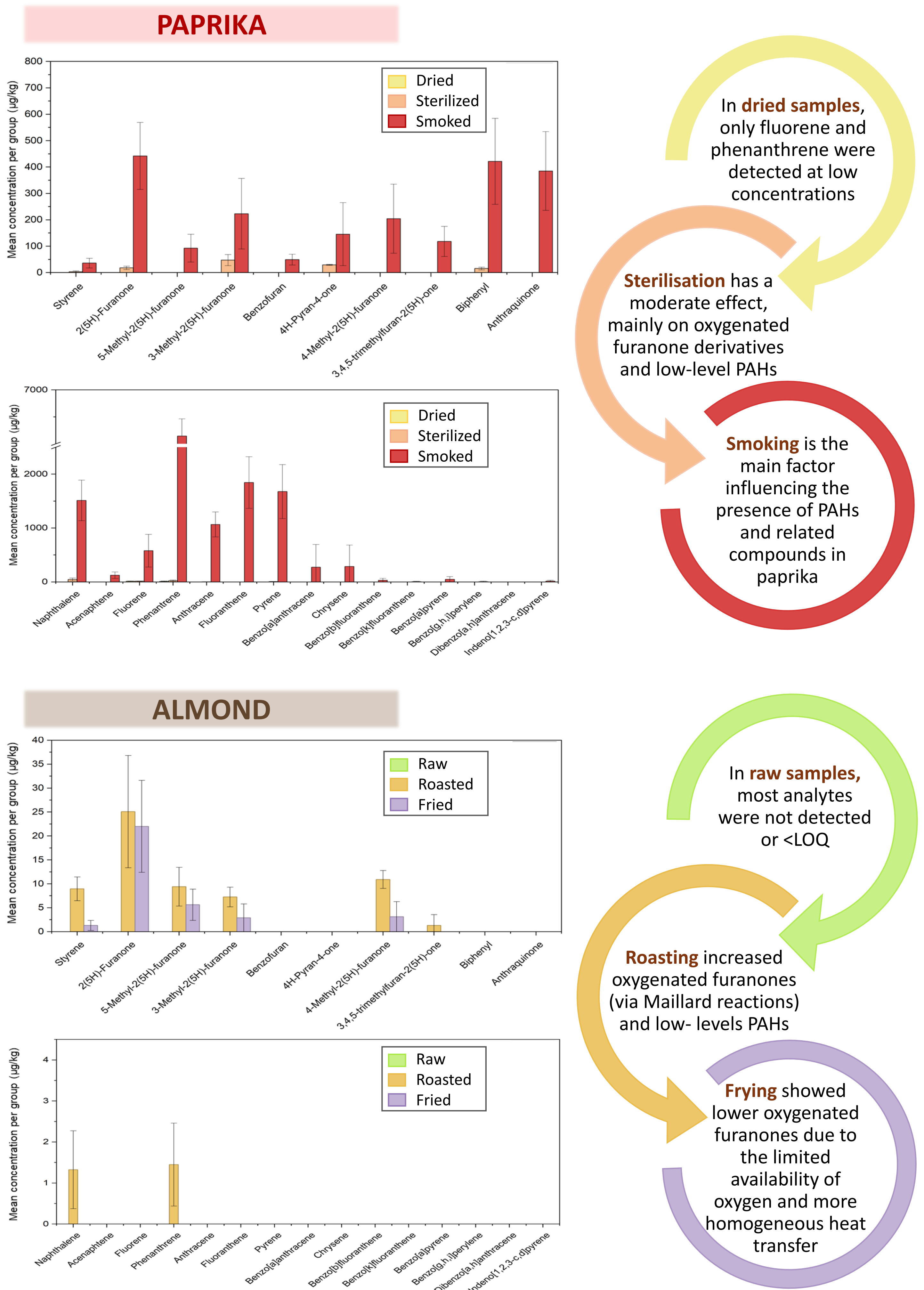


VALIDATION

PAPRIKA					
Family compounds	Linearity (R ²)	ME (%)	Spiking levels (µg/kg)	Mean recovery (%)	LOQs (µg/kg)
PAHs	0.999	-46.7 – 4.8	5	92.9 – 102.9	5
			50	93.0 – 105.8	
Furanones	0.999	3.8 – 41.7	5	97.0 – 102.6	5
			50	98.7 – 104.6	
Other emerging compounds	0.999	-20.0 – 30.0	5	88.6 – 102.6	5
			50	101.7 – 106.1	

ALMOND					
Family compounds	Linearity (R ²)	ME (%)	Spiking levels (µg/kg)	Mean recovery (%)	LOQs (µg/kg)
PAHs	0.999	-46.7 – 4.8	5	92.9 – 102.9	5
			50	93.0 – 105.8	
Furanones	0.999	3.8 – 41.7	5	97.0 – 102.6	5
			50	98.7 – 104.6	
Other emerging compounds	0.999	-20.0 – 30.0	5	88.6 – 102.6	5
			50	101.7 – 106.1	

RESULTS



CONCLUSION

- ✓ A robust **UAE-GC-Orbitrap-HRMS** method was developed and validated for the simultaneous determination of 25 thermal process-induced compounds (both priority and emerging) in complex matrices.
- ✓ The LOQ was set at **5 µg/kg** for all target analytes in both paprika and almond.
- ✓ Dried and sterilised paprika samples showed low PAHs levels, in some cases below the LOQ, smoked paprika exhibited markedly higher concentrations, with **benzofuran** and **anthraquinone** serving as markers of **wood combustion**.
- ✓ In almond, thermal processing primarily promoted the formation of **furanone derivatives**, particularly during **roasting**, while PAH levels remained negligible.

REFERENCES: [1] A. M. Navarro-Herrera, A. Rivera-Pérez, y A. Garrido Frenich. (2026). A novel approach based on GC-Orbitrap-HRMS analysis for the simultaneous multi-class determination of priority and emerging contaminants in thermally processed paprika and almond. *Food Control*, 189, 112340.